

Experimental Contact IV: The Decoherence Floor and the Equivalence Principle

Forbidden Collapse, the Dilation Floor, and the Two Scaling Laws of Locked Matter

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Abstract

This note closes the remaining phenomenological obligations of the FRC experimental programme. *The decoherence floor (T6)*: below registration the framework is exactly unitary, so an isolated superposition keeps full fringe visibility at any mass, forever — FRC *forbids* spontaneous-collapse decoherence of the GRW/CSL type, with no parameter to tune; every matter-wave interference record, including the recent 1.7×10^5 amu sodium-cluster result, is a passed test, and any confirmed anomalous decoherence in uniform potential falsifies the framework outright. The only intrinsic floor is gravitational and is inherited parameter-free from the corpus readings: mass is winding rate, so a composite’s internal clocks read branch proper time, and a proper-time differential between interferometer arms writes which-path information into the internal state — gravitational time-dilation decoherence, with the visibility given exactly by the characteristic function of the internal winding distribution. The finite substrate adds one discriminator: the internal spectrum is integer windings, so visibility *revives exactly* at the cycle recurrence — dephasing, never collapse. *The equivalence principle (T7)*: the gravity companion’s coherent-additivity lemma is given its experiment-facing reading. Two scaling laws coexist: energy (power-additive) observables scale with the full cardinality m for locked and unlocked matter alike — the stationary field of distributed, internally locked clusters sources the entire mass, so gases gravitate fully, and the acceleration of any locked test body is exactly independent of its mass and composition: $\eta = 0$ identically, consistent with MICROSCOPE and atom-interferometric tests and falsifiable by any confirmed violation. The $\sqrt{m}$ of the lemma lives exclusively in amplitude-coherent observables — the synchronisation coupling of m mutually incoherent phases at one site, $\mathbb{E}|\Sigma|^2 = m$ exactly — which yields an FRC-specific differential prediction: the entangling visibility of BMV-type experiments degrades when the source mass’s collective phase is incoherent. All claims are machine-verified by exact arithmetic (`decoherence.py`, `equivalence.py`; nine checks, all passing).

1 T6: the decoherence floor

1.1 Spontaneous collapse is forbidden

Proposition 1.1 (No intrinsic visibility loss). *Below registration, every FRC evolution — the drive and every form-preserving propagator — conserves ledger norms exactly. An isolated superposition therefore retains unit fringe contrast at every drive time:* verified as a ring identity, with the interference weight sweeping the exact extremes 0 and 4 at every evolution time and phase offset (`decoherence.py`, Q1). *FRC admits no spontaneous-collapse channel at any mass scale, and has no parameter with which to acquire one.*

The confrontation is double-edged in the framework’s favour. Collapse models [5, 6] survive by sliding their rate parameters as interference records climb; FRC predicts *exact*

persistence of coherence in uniform potential, so the molecular and nanoparticle interference records — oligoporphyrins beyond 25 kDa [7], sodium clusters beyond 1.7×10^5 amu [8] — are passed tests rather than constraints, while a single statistically confirmed anomalous decoherence with a collapse signature (mass-scaled visibility loss, spontaneous heating) would falsify FRC outright. The forbidden region of the Sorkin note [3] has an exact analogue here: between zero intrinsic decoherence and the gravitational floor below, the framework leaves no room to retreat.

1.2 The gravitational floor: dilation dephasing with exact revivals

Proposition 1.2 (The dilation floor). *Let a locked cluster have internal winding distribution p_E (its thermal internal state, in the corpus reading where energy is winding number), and let its two interferometer arms accumulate proper-time differential τ from the derived potential [2]. Then the fringe visibility is the characteristic function of the internal distribution,*

$$V(\tau) = \left| \sum_E p_E \zeta^{E\tau} \right|, \quad (1)$$

exactly; for a thermal (binomial) distribution this decays with the Gaussian envelope $V \simeq e^{-\sigma^2(\omega\tau)^2/2}$ — gravitational time-dilation decoherence [9], derived here from mass is winding rate with no additional assumption. And because the spectrum is integer windings, the visibility revives exactly at the cycle period: $V(n) = 1$ identically. All three statements machine-verified: the exact ring form over the full sweep, the Gaussian envelope, and the exact recurrence (decoherence.py, Q2–Q3).

Remark 1.3 (The experimental contour). The floor is invisible to every existing record: molecule and nanoparticle interferometers are horizontal (negligible proper-time differential between arms) and fast, so the dilation phase is many orders below threshold. The floor becomes a target where [9] located it — massive thermal objects with vertically separated arms held for long times — which is precisely the regime of the proposed high-mass roadmaps [10]. The FRC-specific discriminator is the revival: a collapse model destroys coherence irreversibly, the dilation floor of Proposition 1.2 returns it, in principle exactly — the same reversibility that distinguishes relational decoherence from dynamical reduction throughout the framework.

2 T7: the equivalence principle and the two scaling laws

The gravity companion’s Lemma 4.2 states that a locked cluster of cardinality m couples to the ambient field with coefficient exactly m , while an unlocked single-site ensemble of m independent phases couples with expected magnitude $O(\sqrt{m})$ [2]. Left without phenomenological commentary, this invites a referee’s objection — does hot matter gravitate less? The resolution is that the lemma describes *amplitude* coupling, and the observables of gravitation are *power* observables.

Proposition 2.1 (Power additivity: full sourcing and the exact EP null). *(i) The stationary bias field is linear in its sources: the field of many separately locked clusters is the exact sum of their full- m contributions — verified by exact rational solution of the lattice Poisson equation, superposition and the m -coefficient both exact on every cell (equivalence.py, E1). A gas of internally locked atoms therefore sources its entire mass: thermal centre-of-mass motion does not unlock atoms, and matter gravitates by its energy. (ii) The pull on a locked test cluster is $m_t \nabla u$ and its inertia is m_t , so its acceleration is ∇u , independent of mass and composition*

identically: $\eta = 0$ exactly (verified, E2). *MICROSCOPE's* $|\eta| \lesssim 10^{-15}$ [11] and the atom-interferometric tests at 10^{-12} [12] are exact nulls for FRC — Class-C survival entries, with any confirmed EP violation a direct falsifier.

Proposition 2.2 (Amplitude coherence: where the $\sqrt{m}$ lives). *For m mutually incoherent phases at a single site, $\mathbb{E}|\sum_i \zeta^{\theta_i}|^2 = m$ exactly (character orthogonality; verified by exhaustive enumeration for small m and by sampling at $m = 10^4$, E3–E4), against m^2 for a locked cluster: the $\sqrt{m}$ suppression affects only amplitude-coherent observables — the synchronisation coupling of a collective phase — and no energy observable anywhere.*

Remark 2.3 (The differential prediction). The two scaling laws yield an FRC-specific experimental handle: in a BMV-type experiment [4], the *entangling* channel is amplitude-coherent, so the entanglement visibility degrades by the coherent fraction when the source mass's collective phase is incoherent, while its gravitational pull (a power observable) is unchanged. Cold, internally rigid source masses entangle; hot or internally incoherent ones gravitate identically but entangle weakly. No semiclassical model and no standard quantisation of gravity makes this particular pairing of predictions — equal force, unequal entanglement — which marks it as the framework's sharpest novel signature in the gravitational sector.

3 Status

Closed. T6: spontaneous collapse forbidden (passed against all interference records, falsifiable by any confirmed anomaly); the dilation floor derived parameter-free with its exact revival discriminator. T7: the EP null exact ($\eta = 0$ identically, MICROSCOPE-consistent); the $\sqrt{m}$ localised to amplitude observables; the equal-force/unequal-entanglement differential prediction filed. Nine machine checks, all passing.

The tier list after this note. T1–T7 closed. Remaining: T8 (the amplitude-granularity ceiling for deep quantum circuits), T9 (qudit emulation of the worked examples), and the Ω -ledger consolidation — the single table on which every bound above must agree.

Reproducibility

The scripts `decoherence.py` and `equivalence.py` reside in the `validation/` folder of the companion draft, slated for <https://github.com/gamayos/frc-quantum-numerics>. Recorded output: nine PASS lines and the terminal lines `decoherence: all checks passed, equivalence: all checks passed`.

References

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